

A/B Testing and Online Experiments Under Diminishing Marginal Effects: Regret Minimization and Statistical Inference

Course Project Report (DBMS)

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Abstract

This report studies the problem of online experimentation in environments where the reward obtained from a strategy exhibits diminishing marginal effects. Traditional A/B testing and classical multi-armed bandit algorithms are primarily designed under the assumption that rewards scale linearly with the amount of traffic allocated to a strategy. Under this framework, the objective is typically to identify a single best alternative and assign all users to it. However, many real-world applications, including online advertising, recommendation systems, and digital marketing platforms, violate this assumption due to budget constraints, user fatigue, and saturation effects.

The report reviews the theoretical framework proposed for online experimentation under diminishing marginal effects, where the objective shifts from selecting a single optimal strategy to determining an optimal allocation of traffic across multiple strategies. Key concepts such as regret minimization, statistical inference, and the Central Limit Theorem are discussed in this context. The paper establishes fundamental regret lower bounds, introduces algorithms for near-optimal traffic allocation, and develops statistical guarantees for inference on the optimal reward. By jointly addressing optimization and inference, the framework provides a principled approach for large-scale online decision-making systems operating under realistic nonlinear reward structures.

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1 Introduction

Online platforms such as e-commerce websites, social media applications, and digital advertising systems continuously conduct A/B tests and online experiments to evaluate alternative strategies and improve user engagement. Traditional A/B testing assumes that the effect of a strategy scales linearly with the amount of traffic allocated to it. Under this assumption, if one strategy performs better than another, the optimal decision is simply to identify the better strategy and assign all users to it. Consequently, most classical A/B testing and multi-armed bandit methods focus on selecting a single best strategy.

However, the assumption of linearity is often unrealistic in large-scale online systems. In practice, many applications exhibit *diminishing marginal effects*, where the incremental benefit of allocating additional users to a strategy decreases as its traffic share increases. For example, in online advertising systems, advertisers may have limited budgets, causing the effectiveness of an advertising strategy to decline as more impressions are delivered. Similarly, repeated exposure to the same recommendation or advertisement can lead to user fatigue or *banner blindness*, reducing its marginal impact. In such scenarios, allocating all traffic to a single strategy may no longer be optimal. Instead, the highest expected reward may be achieved through a mixture of strategies, where only a fraction of users are assigned to each alternative.

Existing approaches are not designed for this setting. Classical A/B testing methods primarily focus on statistical inference, while multi-armed bandit algorithms emphasize regret minimization and efficient learning. When diminishing marginal effects are present, the objective becomes finding the optimal traffic allocation rather than selecting a single winning strategy. Furthermore, even algorithms that successfully minimize regret may fail to provide reliable statistical inference, making it difficult to construct valid confidence intervals or perform hypothesis testing for the optimal reward.

This paper addresses both challenges simultaneously. The authors formulate online experimentation under diminishing marginal effects as an optimization problem over traffic allocation proportions. They first establish a theoretical lower bound on achievable regret and propose a bisection-based algorithm that achieves near-optimal regret. They then prove an impossibility result showing that regret-optimal algorithms alone cannot guarantee valid statistical inference. To overcome this limitation, the paper introduces a novel zero-order gradient-based algorithm that exploits higher-order smoothness assumptions. The proposed method achieves near-optimal regret of order $O(\sqrt{T} \log T)$ while also establishing a valid Central Limit Theorem (CLT) for the estimator of the optimal reward, enabling reliable statistical inference.

The contributions of this work are significant for modern data-driven decision-making systems. By jointly addressing optimization and inference in the presence of diminishing marginal effects, the paper provides a framework that is both theoretically rigorous and practically relevant for large-scale online experimentation.

2 Background and Preliminaries

This section introduces the fundamental concepts required to understand the problem of online experimentation under diminishing marginal effects. In particular, we review classical A/B testing, multi-armed bandit problems, regret minimization, statistical inference through the Central Limit Theorem (CLT), and the notion of diminishing marginal effects.

2.1 A/B Testing

A/B testing is one of the most widely used methodologies for data-driven decision making in online systems. In a typical A/B test, users are randomly divided into two groups and exposed to two alternative strategies, denoted as Strategy A and Strategy B. The performance of each strategy is then evaluated using a predefined metric such as click-through rate (CTR), conversion rate, user engagement, or revenue.

The primary objective of an A/B test is to determine whether one strategy performs significantly better than the other. Classical A/B testing assumes that the effect of a strategy scales linearly with the amount of traffic allocated to it. Under this assumption, once a superior strategy is identified, all traffic is eventually assigned to that strategy.

2.2 Multi-Armed Bandits

The Multi-Armed Bandit (MAB) framework models sequential decision-making under uncertainty. The name originates from the analogy of a gambler choosing among multiple slot machines (arms), each with an unknown reward distribution.

At each time step, the decision maker selects an arm and observes a stochastic reward. The fundamental challenge is balancing:

- **Exploration:** Collecting information about uncertain alternatives.
- **Exploitation:** Using the currently best-known alternative to maximize reward.

Unlike classical A/B testing, which primarily focuses on statistical significance, bandit algorithms emphasize efficient learning and cumulative reward maximization. However, standard bandit formulations generally assume that one arm is globally optimal and therefore aim to identify a single best strategy.

2.3 Regret Minimization

A central performance measure in online learning and bandit problems is *regret*. Regret quantifies the cumulative loss incurred due to not knowing the optimal decision from the beginning.

Let x_t denote the action selected at time period t and let x^* denote the optimal action. If $f(x)$ represents the expected reward associated with action x , then the cumulative regret over a time horizon T is defined as

$$R_T = \mathbb{E} \left[\sum_{t=1}^T (f(x^*) - f(x_t)) \right].$$

A learning algorithm is considered effective if its regret grows sublinearly with T . In such cases,

$$\frac{R_T}{T} \rightarrow 0,$$

implying that the average loss per time period vanishes as more data are collected.

2.4 Central Limit Theorem and Statistical Inference

In addition to optimization, online experimentation often requires reliable statistical inference. Statistical inference enables practitioners to construct confidence intervals, perform hypothesis tests, estimate effect sizes, and quantify uncertainty.

A fundamental result supporting statistical inference is the Central Limit Theorem (CLT). Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables with mean μ and variance σ^2 . Then

$$\sqrt{n} (\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2),$$

where $\bar{X}$ denotes the sample mean.

The CLT implies that, for sufficiently large sample sizes, estimators are approximately normally distributed. Consequently, confidence intervals and hypothesis tests can be constructed using standard Gaussian approximations. A major objective of the paper under study is to establish a valid CLT for the estimator of the optimal reward value.

2.5 Diminishing Marginal Effects

Many real-world systems exhibit *diminishing marginal effects*, meaning that the incremental benefit obtained from assigning additional traffic to a strategy decreases as its usage increases.

Mathematically, if $r(x)$ denotes the expected reward when a fraction x of traffic is assigned to a strategy, then diminishing marginal effects correspond to

$$r'(x) > 0, \quad r''(x) < 0.$$

The first condition indicates that allocating more traffic generally increases reward, while the second condition implies that the rate of increase gradually decreases. Such reward functions are therefore concave.

Examples of diminishing marginal effects include:

- **Advertising Systems:** Advertisers often have limited budgets. As more impressions are delivered, the effectiveness of additional impressions decreases.

- **Recommendation Systems:** Repeated exposure to similar recommendations may reduce user engagement due to recommendation fatigue.
- **Online Advertising:** Users repeatedly exposed to the same advertisement may develop banner blindness, lowering the marginal effectiveness of future exposures.

The presence of diminishing marginal effects fundamentally changes the objective of online experimentation. Instead of identifying a single winning strategy, the optimal solution may involve allocating traffic between multiple strategies. Consequently, the optimal allocation may occur at an interior point

$$0 < x^* < 1,$$

rather than assigning all traffic to a single strategy. This observation forms the primary motivation for the problem studied in this report.

3 Problem Formulation

The paper studies online experimentation in the presence of diminishing marginal effects. Consider two competing strategies, labeled Strategy 1 and Strategy 2. These strategies may represent two recommendation algorithms, two advertising policies, or the control and treatment groups in a classical A/B test.

The experiment is conducted over a finite horizon of T time periods. At each time period t , the decision maker chooses a traffic allocation proportion

$$x_t \in [0, 1],$$

where x_t represents the fraction of users assigned to Strategy 1 and $1 - x_t$ represents the fraction assigned to Strategy 2. Users are assigned randomly according to these proportions.

3.1 Reward Model

To model diminishing marginal effects, the paper assumes that each strategy possesses an unknown reward function.

Let

$$r_1 : [0, 1] \rightarrow \mathbb{R}$$

denote the expected reward obtained from Strategy 1 when a fraction x of users is assigned to it. Similarly,

$$r_2 : [0, 1] \rightarrow \mathbb{R}$$

denotes the expected reward obtained from Strategy 2 when a fraction $1 - x$ of users is assigned to it.

Unlike classical A/B testing, where rewards are typically assumed to be linear in the allocated traffic, the paper assumes that both reward functions are non-decreasing and concave. This captures the phenomenon of diminishing marginal effects, where additional traffic contributes progressively smaller increases in reward.

The total expected reward obtained from one period is therefore

$$f(x) = r_1(x) + r_2(1 - x).$$

Since both $r_1(\cdot)$ and $r_2(\cdot)$ are concave, the function $f(\cdot)$ is also concave.

3.2 Stochastic Observations

The true reward functions are unknown and cannot be observed directly. Instead, the decision maker observes noisy rewards.

For Strategy 1, the observed reward is

$$Y_{1,t}(x_t) = r_1(x_t) + \xi_{1,t}(x_t),$$

where $\xi_{1,t}(x_t)$ is a random noise term with mean zero.

Similarly, for Strategy 2,

$$Y_{2,t}(1 - x_t) = r_2(1 - x_t) + \xi_{2,t}(1 - x_t),$$

where $\xi_{2,t}(1 - x_t)$ is another mean-zero random variable.

The paper assumes that these noise variables have bounded variance, i.e.,

$$\mathbb{E} [\xi_i^2(x) \mid x] \leq \sigma^2, \quad i = 1, 2.$$

Consequently, the observed rewards are stochastic realizations of the underlying reward functions.

3.3 Optimal Traffic Allocation

The objective is to find the traffic allocation that maximizes the expected reward.

Define

$$x^* = \arg \max_{x \in [0,1]} f(x).$$

The quantity x^* represents the optimal proportion of users assigned to Strategy 1.

In classical A/B testing with linear reward functions, the optimal solution typically occurs at one of the boundaries, namely $x^* = 0$ or $x^* = 1$, implying that all users should

eventually be assigned to a single strategy. However, under diminishing marginal effects, the optimal allocation may lie in the interior of the interval,

$$0 < x^* < 1,$$

indicating that a mixture of both strategies yields the highest expected reward.

3.4 Objectives of the Experiment

The paper considers two fundamental objectives.

3.4.1 Regret Minimization

The first objective is to design an adaptive policy that learns the optimal allocation while minimizing cumulative regret.

Let x_t denote the allocation chosen at time period t . The cumulative regret of a policy π is defined as

$$R_T(\pi) = \mathbb{E} \left[\sum_{t=1}^T (f(x^*) - f(x_t)) \right].$$

Regret measures the total loss incurred due to not knowing the optimal allocation from the beginning. A desirable algorithm should achieve sublinear regret growth, ensuring that the average regret per period approaches zero as T increases.

3.4.2 Statistical Inference

The second objective is to estimate the optimal reward value

$$f(x^*)$$

and provide valid statistical inference for this quantity.

In practical online experimentation, decision makers often require confidence intervals, hypothesis tests, and estimates of minimum detectable effects before deploying a strategy. Therefore, it is not sufficient to merely identify a good allocation; one must also quantify the uncertainty associated with the estimated optimal reward.

The central challenge addressed by the paper is the simultaneous achievement of both objectives: near-optimal regret minimization and statistically valid inference.

4 Regret Minimization Framework

The first objective considered in the paper is regret minimization. Given a finite time horizon T , the goal is to design an adaptive allocation policy that learns the optimal

traffic allocation while minimizing cumulative regret. Before presenting the proposed algorithm, the paper establishes a fundamental lower bound on the achievable regret and characterizes the class of reward functions under consideration.

4.1 Function Class and Smoothness Assumptions

Since the overall reward function

$$f(x) = r_1(x) + r_2(1 - x)$$

is concave, the optimization problem can be viewed as a stochastic concave optimization problem over the interval $[0, 1]$. To analyze the performance of learning algorithms, the paper introduces a smoothness class of functions denoted by $\mathcal{F}_\beta(L)$.

For a smoothness parameter $\beta > 0$, let $\ell = \lfloor \beta \rfloor$ be the largest integer strictly smaller than β . A function belongs to $\mathcal{F}_\beta(L)$ if it satisfies the Hölder-type smoothness condition for all $x, z \in [0, 1]$:

$$\left| f(z) - \sum_{0 \leq m \leq \ell} \frac{1}{m!} f^{(m)}(x) (z - x)^m \right| \leq L |z - x|^\beta.$$

Intuitively, larger values of β correspond to smoother functions possessing higher-order derivatives, where the function is well-approximated by its Taylor polynomial.

The paper further considers the subclass $\mathcal{F}_{\alpha,\beta}(L)$, which is a subset of $\mathcal{F}_\beta(L)$ defined as:

$$\mathcal{F}_{\alpha,\beta}(L) = \{f \in \mathcal{F}_\beta(L) : f \text{ is } \alpha\text{-strongly concave}\}.$$

Strong concavity plays an important role in optimization. It guarantees the existence of a unique maximizer

$$x^* = \arg \max_{x \in [0,1]} f(x),$$

and ensures that the reward decreases quadratically as one moves away from the optimum. Consequently, estimating the location of x^* becomes equivalent to estimating the optimal reward value.

4.2 Regret Lower Bound

A natural question is whether increasingly sophisticated algorithms can drive the cumulative regret to arbitrarily small values. The paper answers this question negatively by establishing a minimax lower bound.

Theorem 2.1 (Regret Lower Bound). *Suppose the reward function belongs to the class $\mathcal{F}_{\alpha,\beta}(L)$ with $\beta \geq 2$, and assume that the observation noise is Gaussian with non-vanishing variance. Then, for any adaptive policy π ,*

$$\sup_{f \in \mathcal{F}_{\alpha,\beta}(L)} R_T(\pi) \geq C\sqrt{T},$$

where $C > 0$ is a constant independent of T .

This theorem establishes that no algorithm can achieve cumulative regret growing asymptotically slower than order $\sqrt{T}$. In other words, regardless of the learning strategy employed, there exists a problem instance on which the regret is at least proportional to $\sqrt{T}$.

The result is particularly noteworthy because it remains valid even when the reward function possesses higher-order smoothness. One might expect that smoother functions would be significantly easier to optimize; however, the theorem shows that additional smoothness alone cannot improve the fundamental regret rate.

The lower bound serves as a benchmark for evaluating all subsequent algorithms. Any algorithm achieving regret close to

$$O(\sqrt{T})$$

can be regarded as near-optimal from a minimax perspective. Consequently, the objective of the paper becomes designing an algorithm whose regret matches this lower bound as closely as possible while simultaneously enabling valid statistical inference.

4.3 Bisection Search Algorithm

After establishing the regret lower bound, the paper proposes a bisection-based search procedure that achieves a nearly matching upper bound. The algorithm exploits the concavity of the reward function in order to efficiently locate the optimal traffic allocation x^* .

The key observation is that for a concave function, local information about the slope of the function provides information about the location of the global optimum. If the derivative is positive at a point, the optimum must lie to the right. Conversely, if the derivative is negative, the optimum must lie to the left. This property allows the search interval containing the optimizer to be repeatedly reduced.

4.3.1 Interval Shrinking Procedure

The algorithm begins with the full feasible interval

$$I_0 = [0, 1].$$

At each stage, the current interval is examined and statistical tests are performed to determine whether the optimizer lies in the left half, the right half, or remains uncertain. Using noisy observations of the reward function, the algorithm constructs a sequence of shrinking intervals

$$I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots.$$

Each iteration reduces the uncertainty regarding the location of the optimal allocation. Due to the concavity of the reward function, the true optimizer remains inside the interval with high probability.

The number of samples allocated to each shrinking stage k is carefully chosen as

$$N_k = \left\lceil \left(\frac{3}{2}\right)^{4k} \log^2(T) \right\rceil,$$

which ensures that the probability of incorrectly discarding the interval containing the optimizer becomes sufficiently small.

4.3.2 Boundary Detection

An important feature of the algorithm is its ability to determine whether the optimum lies at the boundary of the feasible region. After a predetermined number of shrinking iterations τ_1 , the resulting interval

$$I_{\tau_1} = [x_{L,\tau_1}, x_{R,\tau_1}]$$

is examined. Three cases may occur:

1. $x_{L,\tau_1} > 0$ and $x_{R,\tau_1} < 1$: In this case, the optimizer is likely to lie in the interior of the interval, and further shrinking is performed.
2. $x_{L,\tau_1} = 0$: The optimizer is likely located at the left boundary. The algorithm subsequently allocates all traffic to Strategy 2.
3. $x_{R,\tau_1} = 1$: The optimizer is likely located at the right boundary. The algorithm subsequently allocates all traffic to Strategy 1.

This boundary detection mechanism is particularly important because the optimal allocation under diminishing marginal effects is not guaranteed to lie in the interior of the feasible region.

4.3.3 Final Allocation Strategy

If the optimizer is determined to lie in the interior, the interval shrinking procedure continues until a sufficiently small interval

$$I_{\tau_2} = [x_{L,\tau_2}, x_{R,\tau_2}]$$

is obtained. The final traffic allocation is chosen as the midpoint

$$\hat{x} = \frac{x_{L,\tau_2} + x_{R,\tau_2}}{2}.$$

The algorithm then applies this allocation for the remainder of the experimental horizon. The observed rewards collected during this stage are also used to estimate the optimal reward value.

4.3.4 Regret Guarantee

The performance of the bisection search algorithm is summarized by Theorem 2.2.

Theorem 2.2. *Under the assumptions of strong concavity and sub-Gaussian observation noise, the cumulative regret of the proposed policy π satisfies*

$$R_T(\pi) = O\left(\sqrt{T} \log T\right).$$

This upper bound nearly matches the minimax lower bound established in Theorem 2.1,

$$R_T(\pi) = \Omega\left(\sqrt{T}\right),$$

differing only by a logarithmic factor. Therefore, the proposed algorithm is essentially optimal from a regret minimization perspective.

The result demonstrates that the optimizer can be located efficiently despite noisy observations and unknown reward functions. Consequently, the algorithm successfully solves the first objective of the paper, namely regret minimization. However, as shown in the next section, achieving low regret alone is insufficient for reliable statistical inference.

5 Challenges in Statistical Inference and an Impossibility Result

The previous section demonstrated that the bisection search algorithm achieves near-optimal regret. From an optimization perspective, this appears to solve the problem of online experimentation under diminishing marginal effects. However, practical online experiments require more than merely identifying a near-optimal traffic allocation. In many applications, decision makers also require reliable statistical inference regarding the optimal reward value.

Examples of such inference tasks include the construction of confidence intervals, hypothesis testing, power analysis, significance testing, and estimation of minimum detectable effects. These quantities are essential before deploying a strategy in production, since they provide a quantitative measure of uncertainty and expected business impact.

5.1 The Need for a Valid Central Limit Theorem

Let $f(x^*)$ denote the optimal expected reward. In addition to learning the optimizer x^* , the experimenter wishes to estimate this quantity and construct confidence intervals around it.

In classical statistical inference, confidence intervals are typically derived from a Central Limit Theorem (CLT) of the form

$$\sqrt{T}\left(\hat{f}_T - f(x^*)\right) \xrightarrow{d} N(0, \sigma^2),$$

where $\hat{f}_T$ is an estimator of the optimal reward.

A crucial requirement for such a result is that the asymptotic bias of the estimator vanishes. If the limiting distribution is centered away from zero, then confidence intervals constructed using the CLT become systematically incorrect. Consequently, establishing a valid CLT is essential for reliable statistical inference.

5.2 Why Regret Minimization Alone is Insufficient

A natural question is whether the bisection search algorithm introduced in the previous section already provides the desired statistical guarantees. The paper demonstrates that the answer is negative. Although Algorithm 1 achieves nearly optimal regret, it does not produce an estimator whose asymptotic bias vanishes sufficiently quickly.

To illustrate this phenomenon, the authors consider a simple quadratic example and study the empirical distribution of

$$\sqrt{T} \left(\hat{\mu}_f^{(T)} - f(x^*) \right),$$

where $\hat{\mu}_f^{(T)}$ denotes the estimator produced by the bisection algorithm.

The resulting histograms, presented in Figure 1 of the paper, reveal that the distribution remains shifted away from zero even for very large values of T . This indicates the presence of a persistent bias that does not disappear asymptotically. Therefore, despite its excellent regret performance, Algorithm 1 cannot provide reliable confidence intervals for the optimal reward.

5.3 Sources of Difficulty

The paper identifies three major obstacles to constructing a valid CLT:

1. The optimal allocation x^* is unknown and must itself be estimated.
2. The reward function is observed only through noisy stochastic observations.
3. The asymptotic distribution of the estimator must be shown to converge to a Gaussian distribution with zero mean.

Each of these challenges introduces additional estimation error, making statistical inference substantially more difficult than regret minimization alone.

5.4 A Fundamental Impossibility Result

The central theoretical contribution of this section is an impossibility theorem demonstrating that regret minimization and statistical inference cannot generally be achieved simultaneously under the assumptions considered so far.

Theorem 3.1 (Impossibility Theorem). *There exists a problem instance such that for every adaptive policy,*

$$\sup_{f \in \mathcal{F}_{\alpha, 2}(L)} \mathbb{E}[f(x^*) - f(x_T)] \geq \frac{C}{\sqrt{T}},$$

for some constant $C > 0$.

This result states that no algorithm can guarantee optimization error that decreases faster than order $1/\sqrt{T}$. Consequently,

$$\sqrt{T}(f(x_T) - f(x^*)) \not\rightarrow 0.$$

The optimization error remains visible even after the standard $\sqrt{T}$ scaling used in Central Limit Theorems.

This observation explains the persistent bias observed in the empirical results for Algorithm 1. Although the optimization error is sufficiently small for regret minimization, it is not small enough to disappear in the asymptotic distribution required for statistical inference.

5.5 Implications

The impossibility result reveals a fundamental tension between learning and inference. An algorithm may be nearly optimal in terms of regret while simultaneously being unsuitable for constructing confidence intervals and performing hypothesis tests.

Therefore, additional assumptions are required if one wishes to achieve both objectives simultaneously. The remainder of the paper shows that by imposing stronger smoothness conditions on the reward function, it becomes possible to design an algorithm that preserves near-optimal regret while also establishing a valid Central Limit Theorem. This observation motivates the development of the zero-order gradient-based algorithm presented in the next section.

6 Statistical Inference Under Higher-Order Smoothness

6.1 Motivation

The previous section established an important impossibility result. Although Algorithm 1 achieves near-optimal regret, it fails to provide a valid Central Limit Theorem (CLT) for inference on the optimal reward value. The fundamental issue is that the optimization error decreases only at order $T^{-1/2}$, which is not sufficiently fast to vanish after the standard $\sqrt{T}$ normalization used in asymptotic inference.

This motivates the search for additional structural assumptions that may enable both regret minimization and valid statistical inference. The key observation of the paper is that stronger smoothness properties of the reward function can improve the convergence rate of the optimizer estimator without improving the minimax regret lower bound. This additional smoothness ultimately makes it possible to eliminate the asymptotic bias responsible for the failure of Algorithm 1.

6.2 Additional Smoothness Assumption

The paper assumes that the reward function belongs to the class

$$f \in \mathcal{F}_{\alpha,\beta}(L),$$

where

$$\beta > 2.$$

Compared with the assumptions used for Algorithm 1, this condition requires the function to possess higher-order smoothness. Intuitively, a larger value of β implies that the function can be approximated more accurately by local polynomial expansions.

An important insight of the paper is that higher-order smoothness does not improve the minimax regret lower bound, which remains of order

$$\Omega(\sqrt{T}).$$

However, it significantly improves the accuracy with which the optimizer x^* can be estimated. This additional accuracy is crucial for obtaining a valid CLT.

6.3 Overview of Algorithm 2

Algorithm 2 combines two stages:

1. A preliminary bisection search phase.
2. A zero-order gradient optimization phase.

The first stage is identical in spirit to Algorithm 1 and serves to determine whether the optimizer lies near the boundary of the feasible interval.

If the optimizer is believed to lie at the boundary, the algorithm immediately commits to that boundary point.

Otherwise, the algorithm enters a second stage based on projected stochastic gradient descent. Since direct gradient observations are unavailable, the gradient must be estimated using only noisy function evaluations. This setting is known as zero-order optimization.

6.4 Zero-Order Gradient Estimation

Unlike classical gradient methods, the derivative of the reward function is not directly observable. The decision maker only observes noisy realizations of the reward function. Consequently, Algorithm 2 employs a *zero-order* gradient estimation procedure based on randomized finite differences.

At iteration i , independent random variables

$$U_{i,j} \sim \text{Uniform}[-1, 1]$$

are generated for

$$j = 1, \dots, \tilde{N}_i.$$

For each perturbation, the function is evaluated at the two points

$$x_i + h_i U_{i,j}$$

and

$$x_i - h_i U_{i,j},$$

where

$$h_i = i^{-1/2}$$

is a shrinking bandwidth parameter.

Denote the corresponding noisy observations by

$$\hat{f}_{i,j}$$

and

$$\hat{f}'_{i,j}.$$

The gradient estimator is then constructed as

$$\hat{g}_{i,j} = \frac{(\hat{f}_{i,j} - \hat{f}'_{i,j})K(U_{i,j})}{2h_i},$$

where $K(\cdot)$ is a kernel function satisfying

$$\int K(u) du = 0,$$

$$\int uK(u) du = 1,$$

and

$$\int u^m K(u) du = 0, \quad m = 2, \dots, \ell.$$

The final gradient estimate is obtained by averaging over all perturbations,

$$H_i(x_i) = \frac{1}{\tilde{N}_i} \sum_{j=1}^{\tilde{N}_i} \hat{g}_{i,j}.$$

This estimator approximates the derivative of the reward function using only function evaluations and therefore allows gradient-based optimization even when direct gradient information is unavailable.

6.5 Regret Analysis

The first theoretical result establishes that the improved algorithm retains near-optimal regret performance.

Theorem 4.1. Under Assumptions 2.2 and 4.1,

$$R_T(\pi) = O(\sqrt{T} \log T).$$

This matches the minimax lower bound up to a logarithmic factor and therefore preserves the optimal regret guarantees obtained by Algorithm 1.

6.6 Statistical Inference Guarantee

A crucial intermediate result is Lemma 4.2, which shows that the optimizer estimator converges at rate

$$E[|\hat{x}_T - x^*|^2] = O\left(T^{-\frac{\beta-1}{\beta}}\right).$$

Because

$$\frac{\beta-1}{\beta} > \frac{1}{2}$$

whenever $\beta > 2$, the optimization error decreases faster than the scale required for asymptotic inference.

As a consequence, the optimization bias becomes negligible after multiplication by $\sqrt{T}$.

The main inference result is stated in Theorem 4.3.

Theorem 4.3.

Under the assumptions of Theorem 4.1,

$$\sqrt{T} \left(\hat{\mu}_f^{(T)} - f(x^*) \right) \xrightarrow{d} N \left(0, \frac{\text{Var}(\xi_1(x^*) + \xi_2(1 - x^*))}{1 - \gamma_1 - \gamma_2} \right).$$

This theorem establishes a valid Central Limit Theorem with zero asymptotic bias. Consequently, confidence intervals and hypothesis tests for the optimal reward can be constructed using standard Gaussian approximations.

Therefore, Algorithm 2 successfully achieves both objectives of the paper:

- Near-optimal regret minimization.

- Reliable statistical inference.

7 Experimental Analysis

7.1 Simulation Setup

To empirically evaluate the theoretical results presented in the paper, we implemented the proposed algorithms in Python and performed Monte Carlo simulations.

Unless otherwise stated, the reward function was chosen as

$$f(x) = -(x - 0.6)^2 + 2,$$

which is strongly concave and attains its maximum value

$$f(x^*) = 2$$

at

$$x^* = 0.6.$$

Independent Gaussian noise with variance parameter $\sigma^2 = 0.25$ was added to all reward observations. Each experiment was repeated 500 times to study the empirical distribution of the estimators.

The primary quantities of interest were:

- The normalized estimator

$$\sqrt{T} \left(\hat{\mu}_f^{(T)} - f(x^*) \right),$$

- The cumulative regret

$$R_T = \sum_{t=1}^T (f(x^*) - f(x_t)),$$

- The empirical convergence behaviour of the optimization algorithms.

7.2 Bias of Algorithm 1

The first experiment was designed to investigate the impossibility result presented in Section 6.

Using the interval shrinking procedure of Algorithm 1, we generated 5000 independent realizations of the estimator

$$\sqrt{T} \left(\hat{\mu}_f^{(T)} - f(x^*) \right).$$

Figure 1 presents the resulting histogram.

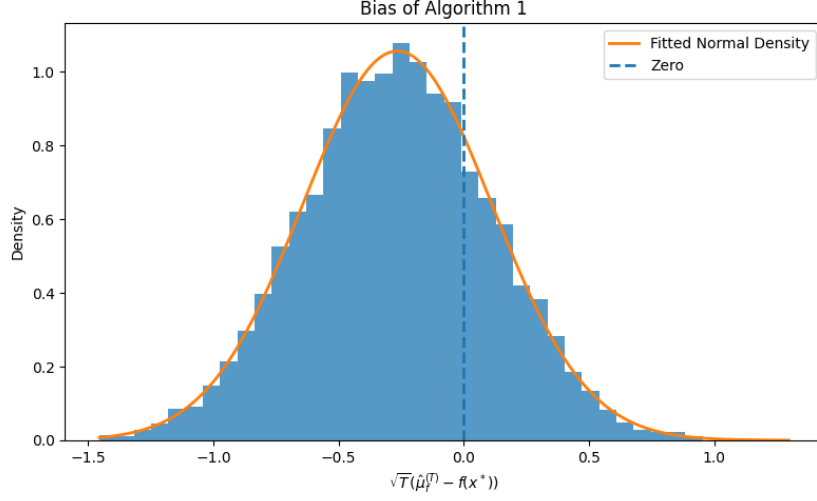


Figure 1: Distribution of the normalized estimator produced by Algorithm 1.

The histogram exhibits a noticeable shift away from zero, indicating the presence of a non-vanishing asymptotic bias. This observation is consistent with the theoretical impossibility result established in Theorem 3.1. Although Algorithm 1 achieves near-optimal regret, it fails to provide a valid Central Limit Theorem for statistical inference.

7.3 Regret Performance of Algorithm 2

The second experiment evaluated the cumulative regret of Algorithm 2.

For different values of the time horizon T , the regret was estimated by averaging over multiple independent runs. Figure 2 shows the resulting regret curve.

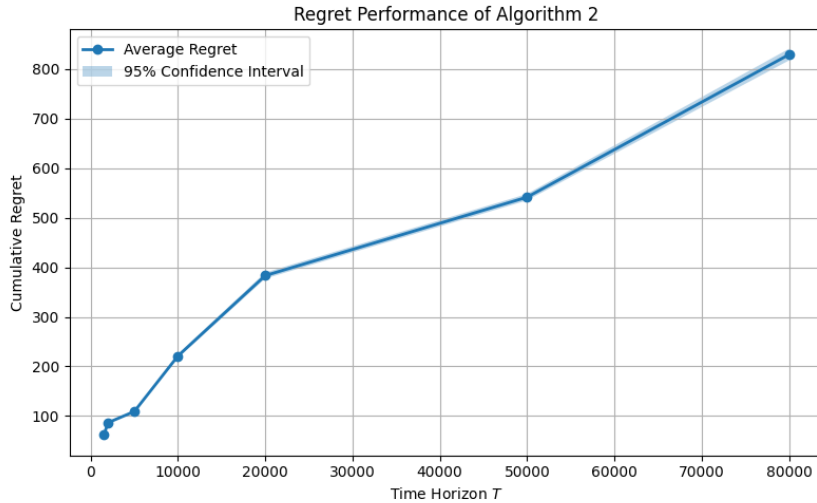


Figure 2: Cumulative regret as a function of the time horizon T .

The empirical results demonstrate a sublinear growth pattern. This agrees with Theorem 4.1, which establishes the regret bound

$$R_T = O(\sqrt{T} \log T).$$

Since the regret grows more slowly than the time horizon itself, the average regret per round decreases toward zero as additional observations are collected.

7.4 Verification of the Central Limit Theorem

To verify the theoretical Central Limit Theorem established in Theorem 4.3, we simulated Algorithm 2 and examined the distribution of

$$\sqrt{T} \left(\hat{\mu}_f^{(T)} - f(x^*) \right).$$

The histogram of the normalized estimator is shown in Figure 3.

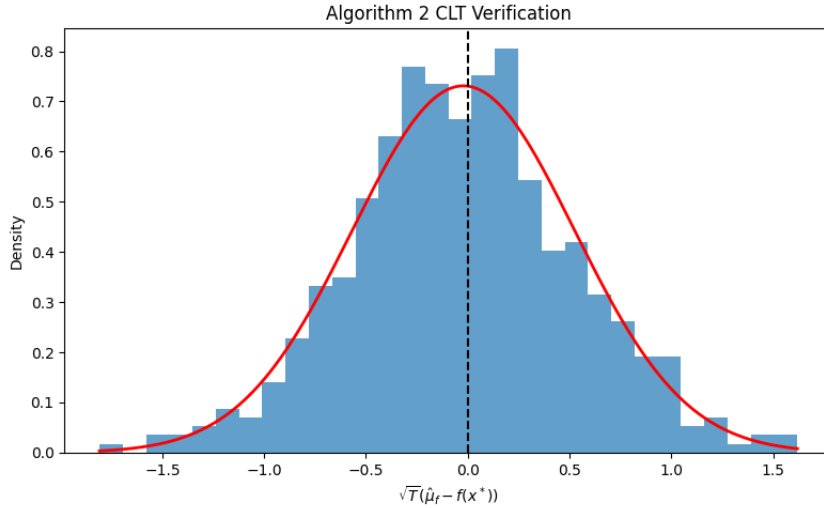


Figure 3: Distribution of the normalized estimator produced by Algorithm 2.

Unlike Algorithm 1, the distribution is approximately centered at zero and exhibits a bell-shaped structure consistent with a Gaussian distribution. This supports the theoretical claim that Algorithm 2 admits a valid Central Limit Theorem with vanishing asymptotic bias.

Consequently, confidence intervals and hypothesis tests based on Algorithm 2 remain statistically valid even while the algorithm simultaneously achieves near-optimal regret.

7.5 Discussion

The experiments support the central message of the paper.

First, achieving near-optimal regret alone does not guarantee reliable statistical inference. Algorithm 1 successfully minimizes regret but produces estimators with persistent asymptotic bias.

Second, higher-order smoothness assumptions enable the design of algorithms that simultaneously achieve optimization and inference objectives. Algorithm 2 preserves the near-optimal regret guarantees of Algorithm 1 while also producing approximately unbiased estimators that satisfy a Central Limit Theorem.

These observations highlight the fundamental trade-off between learning and inference in online experimentation and demonstrate the importance of incorporating statistical considerations into adaptive decision-making algorithms.

8 Possible Extensions

The framework developed in this paper assumes that the reward function is strongly concave. Under this assumption, the optimal traffic allocation is unique and local optimization methods such as bisection search and gradient descent are sufficient for identifying the global optimum. However, real-world online systems may exhibit more complex behavior.

8.1 Extension 1: Non-Concave Reward Functions

In practice, the reward function may not be concave. User interactions, network effects, recommendation diversity, and market dynamics can create multiple local optima in the reward landscape. In such settings, gradient-based methods may converge to a local optimum rather than the globally optimal traffic allocation.

A possible extension is to replace the optimization phase of Algorithm 2 with a global optimization procedure such as Simulated Annealing (SA). Simulated Annealing allows occasional moves to worse solutions with a probability that decreases over time. This enables the algorithm to escape local optima and explore the search space more effectively.

The modified procedure would retain the exploration phase of Algorithm 2 but replace the zero-order gradient descent stage with a Simulated Annealing search over the traffic allocation variable x . Such an approach may be particularly useful when the reward function contains multiple peaks and the concavity assumption is violated.

Although theoretical regret guarantees would become more difficult to establish, Simulated Annealing may provide better practical performance in highly non-convex environments.

8.2 Extension to Non-Concave Reward Functions using Simulated Annealing

The theoretical framework developed in the paper assumes that the total reward function

$$f(x) = r_1(x) + r_2(1 - x)$$

is concave. Under this assumption, the optimization problem possesses a unique global optimum, allowing local search methods such as bisection search and gradient descent to converge reliably.

However, in many real-world applications, the reward landscape may not be concave. User behavior, recommendation diversity, advertiser interactions, and market dynamics can generate multiple local optima. In such situations, gradient-based methods may converge to a locally optimal solution rather than the global optimum.

To illustrate this scenario, consider the reward functions

$$r_1(x) = 1.8 (1 - e^{-4x}) + 0.12 \sin(10\pi x),$$

and

$$r_2(x) = 1.5 (1 - e^{-5x}) + 0.10 \cos(12\pi x).$$

The overall reward function becomes

$$f(x) = r_1(x) + r_2(1 - x),$$

or equivalently,

$$f(x) = 1.8 (1 - e^{-4x}) + 0.12 \sin(10\pi x) + 1.5 (1 - e^{-5(1-x)}) + 0.10 \cos(12\pi(1 - x)).$$

The oscillatory sine and cosine components introduce multiple local maxima into the reward landscape. Consequently, the optimization objective may no longer possess a unique optimum, making local optimization techniques vulnerable to premature convergence.

To address this challenge, a possible extension is to replace the gradient-based optimization stage of Algorithm 2 with a Simulated Annealing (SA) procedure. Simulated Annealing is a global optimization technique that allows occasional transitions to solutions with lower reward values, thereby enabling the algorithm to escape local optima.

Starting from an initial allocation x_t , a candidate solution is generated as

$$x' = x_t + \varepsilon_t,$$

where

$$\varepsilon_t \sim N(0, \sigma_t^2).$$

The candidate is projected onto the feasible region $[0, 1]$ if necessary. Let

$$\Delta = f(x') - f(x_t).$$

If $\Delta > 0$, the candidate solution is accepted. Otherwise, it is accepted with probability

$$P = \exp\left(\frac{\Delta}{T_t}\right),$$

where T_t denotes the temperature parameter. The temperature is gradually decreased according to

$$T_t = T_0 \alpha^t, \quad 0 < \alpha < 1.$$

At high temperatures, the algorithm explores the search space aggressively and can move across valleys separating local maxima. As the temperature decreases, the algorithm gradually shifts toward exploitation and converges to a high-quality solution.

8.2.1 Limitations of the Simulated Annealing Extension

While Simulated Annealing provides a promising approach for handling non-concave reward functions with multiple local optima, it introduces several important limitations compared to the original framework proposed in the paper.

- **Loss of Regret Guarantees:** The original paper establishes the near-optimal regret bound

$$R_T = O\left(\sqrt{T} \log T\right).$$

This guarantee relies heavily on the strong concavity of the reward function and the use of local optimization methods. In contrast, Simulated Annealing intentionally explores suboptimal regions of the search space in order to escape local optima. Consequently, the theoretical regret analysis developed in the paper no longer applies, and no comparable regret bound is currently available for the proposed extension.

- **Increased Computational Complexity:** The bisection search and zeroth-order gradient algorithms considered in the paper possess relatively simple update rules and admit an overall computational complexity of $O(T)$. Simulated Annealing typically requires a large number of candidate evaluations, temperature updates, and exploration steps before convergence. As a result, the computational cost can be significantly higher, especially when the reward function is expensive to evaluate.
- **More Difficult Statistical Inference:** The Central Limit Theorem established in Theorem 4.3 depends on the optimizer converging sufficiently quickly to the true optimum, allowing the final observations to behave approximately as independent samples collected near x^* . Simulated Annealing, however, generates a Markov chain whose successive states are dependent. Therefore, the statistical inference framework developed in the paper cannot be applied directly.
- **Uncertainty in Confidence Interval Construction:** Although Markov Chain Central Limit Theorems may be applicable under suitable ergodicity assumptions, the resulting limiting distribution is determined by the stationary distribution of the Markov chain rather than directly by the optimal reward value $f(x^*)$. Consequently, constructing valid confidence intervals and hypothesis tests for the optimal reward becomes substantially more challenging.

- **Lack of Theoretical Guarantees for Optimality:** In practice, Simulated Annealing often finds near-global optima for highly non-convex problems. However, for finite time horizons there is generally no guarantee that the algorithm has located the true global optimum. The quality of the solution can depend strongly on the temperature schedule, initialization, and exploration parameters.

Despite these limitations, Simulated Annealing remains an attractive extension when the concavity assumption is violated and the reward function contains multiple local maxima. Developing regret guarantees and statistically valid inference procedures for such non-concave online experimentation problems remains an interesting direction for future research.

8.3 Extension 2: Multiple Strategies

The paper focuses exclusively on two competing strategies, corresponding to the classical A/B testing setting. While this assumption simplifies the optimization problem, many real-world online experimentation systems evaluate multiple alternatives simultaneously. Examples include recommendation systems testing several ranking algorithms, advertising platforms comparing multiple bidding strategies, and e-commerce websites evaluating several interface designs.

Suppose there are k competing strategies. Let

$$x_i \geq 0, \quad \sum_{i=1}^k x_i = 1,$$

denote the proportion of traffic allocated to strategy i , where x_i represents the fraction of users assigned to that strategy.

Let $r_i(x_i)$ denote the expected reward generated by strategy i when it receives a fraction x_i of the total traffic. Under diminishing marginal effects, each reward function is assumed to be increasing and concave. The overall reward function becomes

$$f(x_1, \dots, x_k) = \sum_{i=1}^k r_i(x_i).$$

The optimization problem can therefore be formulated as

$$\max_{x_1, \dots, x_k} \sum_{i=1}^k r_i(x_i)$$

subject to

$$x_i \geq 0, \quad \sum_{i=1}^k x_i = 1.$$

Unlike the original problem, which involves a one-dimensional decision variable x , the feasible region is now a $(k - 1)$ -dimensional simplex. Consequently, the bisection search procedure proposed in Algorithm 1 can no longer be applied directly.

A natural extension of Algorithm 2 is to replace the one-dimensional zeroth-order gradient method with a projected zeroth-order gradient descent algorithm on the simplex. After each gradient update, the traffic allocation vector is projected back onto the feasible region

$$\Delta_k = \left\{ (x_1, \dots, x_k) : x_i \geq 0, \sum_{i=1}^k x_i = 1 \right\}.$$

The update rule becomes

$$\mathbf{x}_{t+1} = \Pi_{\Delta_k} \left(\mathbf{x}_t + \eta_t \widehat{\nabla} f(\mathbf{x}_t) \right),$$

where Π_{Δ_k} denotes projection onto the simplex and $\widehat{\nabla} f$ is a zeroth-order gradient estimator constructed from noisy reward observations.

Such an extension would allow the framework to handle modern multi-variant on-line experiments while preserving the fundamental idea of learning an optimal traffic allocation under diminishing marginal effects.

8.3.1 Limitations of the Multi-Strategy Extension

- **Higher-Dimensional Optimization.**

The optimization problem becomes significantly more difficult as the number of strategies increases. Instead of searching over a one-dimensional interval, the algorithm must optimize over a $((k-1))$ -dimensional simplex.

- **Curse of Dimensionality.**

The number of samples required to accurately estimate gradients generally increases with the dimension of the decision space. Consequently, the learning process may become substantially slower for large values of (k) .

- **More Complex Statistical Inference.**

Constructing confidence intervals for the optimal reward becomes more challenging because uncertainty must be propagated through a higher-dimensional optimization procedure.

- **Computational Cost.**

Projection onto the simplex and estimation of multidimensional gradients introduce additional computational overhead compared to the two-strategy setting considered in the paper.

9 Conclusion

This report studied the paper *A/B Test and Online Experiment Under Diminishing Marginal Effects: Regret Minimization and Statistical Inference*. Unlike classical A/B testing and multi-armed bandit formulations, the paper considers situations where the reward obtained from a strategy exhibits diminishing marginal effects. In such settings, allocating all traffic to a single strategy may no longer be optimal, and the objective becomes finding the best traffic allocation between competing strategies.

The paper addresses two fundamental objectives simultaneously: regret minimization and statistical inference. The authors first establish a lower bound of $\Omega(\sqrt{T})$ on the achievable regret and propose a bisection-based algorithm that attains a matching upper bound up to logarithmic factors. They then demonstrate an important impossibility result, showing that regret-optimal algorithms generally fail to provide valid statistical inference due to persistent asymptotic bias.

To overcome this limitation, the paper introduces a zeroth-order gradient-based algorithm under stronger smoothness assumptions. The proposed method achieves near-optimal regret while also establishing a valid Central Limit Theorem for the estimator of the optimal reward. Experimental results further support the theoretical findings by demonstrating both sublinear regret growth and asymptotically unbiased estimation.

In addition to reviewing the original work, two possible extensions were discussed. First, Simulated Annealing was proposed as an alternative optimization technique for non-concave reward functions containing multiple local optima, where gradient-based methods may fail to locate the global optimum. Second, the framework was generalized from the two-strategy setting to a multi-strategy environment with k competing alternatives, leading to an optimization problem over a probability simplex.

Overall, the paper provides an important contribution to the theory of online experimentation by bridging the gap between optimization and statistical inference. Its results offer both theoretical insights and practical guidance for designing reliable online experiments in modern data-driven systems.

Source Code Repository

To ensure reproducibility of the experiments and figures presented in this report, all source code, generated plots, and supplementary materials have been made publicly available through the following GitHub repository:

[https://github.com/Rahul-pandit-2006/
A-B-Testing-and-Online-Experiments-Under-Diminishing-Marginal-Effects](https://github.com/Rahul-pandit-2006/A-B-Testing-and-Online-Experiments-Under-Diminishing-Marginal-Effects)

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